Alexander Baronsky

B.S. Physics & B.S. Computer Science

Rochester, New York
Mobile: 585-851-1786
Email: professionalbaronsky@gmail.com
LinkedIn: linkedin.com/in/alexander-baronsky
GitHub: github.com/ABaronCode
Personal Website: abaroncode.github.io/

Professional Summary

Motivated problem solver with a background in physics and computational sciences. I am skilled in data-driven analysis and computational mathematics. Passionate about applying technical knowledge to continue to learn and apply myself to new challenges in complex fields of physics.

Education

State University of New York at Albany — December, 2024
Bachelor of Science Physics
Bachelor of Science Computer Science
Minor Mathematics

Monroe Community College — May, 2019
Associates of Science Individual Studies in Science

Coursework

Courses in Physics
Quantum Mechanics, Electromagnetism, Thermodynamics, Classical Mechanics, Advanced Physics Laboratory, Modern Physics, Waves, Bayesian Data Analysis and Signal Processing, Mathematics in Physics, Structure of Matter, University Physics

Courses in Computer Science
Capstone Project, Operating Systems, Principles of Programming, Database Systems, Automata and Formal Languages, Assembly Computer Organization, Discrete Structures, Design Analysis and Algorithms, Data Structures, Programing and Hardware Interface

Courses in Mathematics
Calculus of Several Variables, Discrete Probability, Linear Algebra

Courses in Additional Science
Intermediate Biology, Intermediate Chemistry, Biology Laboratory, Chemistry Laboratory

Electives
Elementary German, Intermediate French, Introduction to Climate Change, Social and Ethical Implications of Computer Science, American National Government, United States History, Science Fiction

Professional Experience

Plug Power — April 2026-Present
Engineer - Technical
I operate and maintain fuel cell test equipment, assist failure analysis, prepare electrode materials, run standardized tests and maintain lab equipment. I additionally develop with Tulip and LabView, making successful interfaces to assist recording and storing data which is exported to Power BI.

Starbucks June 2023-April 2026
Barista

Fast-passed enviroment, required excellent communication, multitasking and teamwork. Upkept up-to-date knowledge on tools and products.

Academic Experience

SUNY Albany Capstone Project January-May, 2024
State University of New York at Albany

Sponsored by Mariya Zheleva's research lab, with the goal of assisting in processing spectrum analysis algorithms. Under supervision and direction from the sponsor. My team and I developed a Python plugin, which utilized the algorithm the sponsor worked with. We integrated this tool into a local instance of the open-source IQEngine project. The plugin provided efficient text annotations and automated placement of visual annotations. Which were located around declared frequency bursts in the displayed geospatial wave spectrogram.

Advanced Physics Laboratory January-May, 2024
State University of New York at Albany

Bragg X-Ray

I completed a single rotation of a NaCl deposit, I recorded glancing angles of the X-rays that passed through. Then used known lattice spacing, path difference and recorded angles. To determine the wavelength of the X-ray for each glancing angle. Then with the mean of the tests I found percent-error. I compared the mean with known values to confirm Bragg's law.

Speed of Light

With an oscillator, mirror, light detector and diode laser. I reflected the laser over increments of 1 meter between distances of 5 - 13 meters into the detector. I took measurements of the phase from the oscillator, then with both phase and the distance travelled. I was then able to calculate measurements to determine the speed of the light from the beam.

Estimating the Area of a Circle

I determined the area of circles on a page, with Bayesian scattering analysis. To do this, rice grains from a height of 1 meter were scattered onto a sheet containing five identical non-overlapping circles. Then over ten trials and I found the test probability of rice landing inside each circle. Then compared the test area of the circles to the actual.

Principles of Programming Compiler Project
State University of New York at Albany January-May, 2024

I created a Java compiler which processed and ran calculations from a simplified version of BASIC. Broken down into three parts: lexer, parser and interpreter. The lexer processed code creating tokens, identified commands/data and stored data within a linked list. The parser recursively processed the linked list, identified nodes and connected commands to computations. Nodes were created and added to a HashMap to account for order of operations. Lastly, the interpreter ran the HashMap and processed calculations.

Volunteer Experience

Telescope Operator & Tour Guide — February-August, 2024-Present
University of Rochester: C.E.K. Mees Observatory

I operate the 24" reflector telescope and direct tours of the observatory. I relay information on the facilities and showcase presentations on topics in astronomy to groups of up to 25 people. I handle set-up, running telescope operations and closing procedures. While observing the night sky, I provide information on different astronomical bodies. In addition, I am trained to track weather conditions to ensure the safe use of the technical equipment. At the conclusion of each tour, I update observation logs which include observed objects, viewing conditions, attendance and clarity of the lens.

Mercury Transit — November, 2019
State University of New York at Albany

Conducted outreach for the physics department. I assisted attendees to the campus observatory, explained the transit and how the lenses were safe to view Mercury while it crossed the sun.

Programming Technologies

Languages

Java, Python, C, C++, SQL, HTML, CSS, JavaScript, Scheme, MatLab

Development Environments

Intellij, CLion, Visual Studio Code, Eclipse

Operating Systems

Linux (Ubuntu), MacOS, Microsoft Windows, iOS

Software & Technical Skills

Labview, LaTeX, Microsoft Office, Git/GitHub, Tulip

Awards

Best Capstone Project: Computer Science(2024), Dean's List(2020), Dollars for Scholars Recipient(2016-2018), Eagle Scout(2015)

Extracurricular

SUNY Albany Physics Club(2019-2024), Astronomy Section, Rochester Academy of Science(2012-2026), Chess Club(2012-2016), Violin, Guitar, Competitive Swimmer(2012-2017)

Leadership

Eagle Scout Project
Mendon Presbyterian Church

Acquired funding, designed and directed construction of an outdoor sidewalk from the church kitchen to parking lot. This project assisted parishioners with mobility issues to access the building. In addition, it provides the church with a safe way to move supplies and tools.